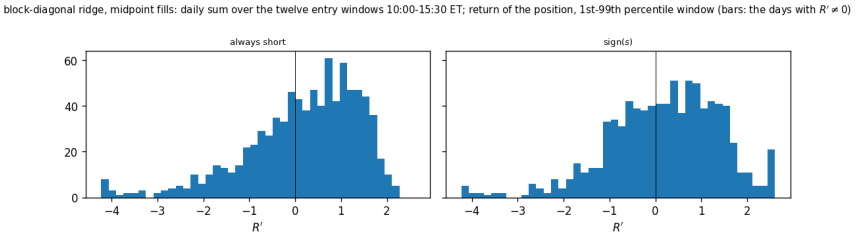
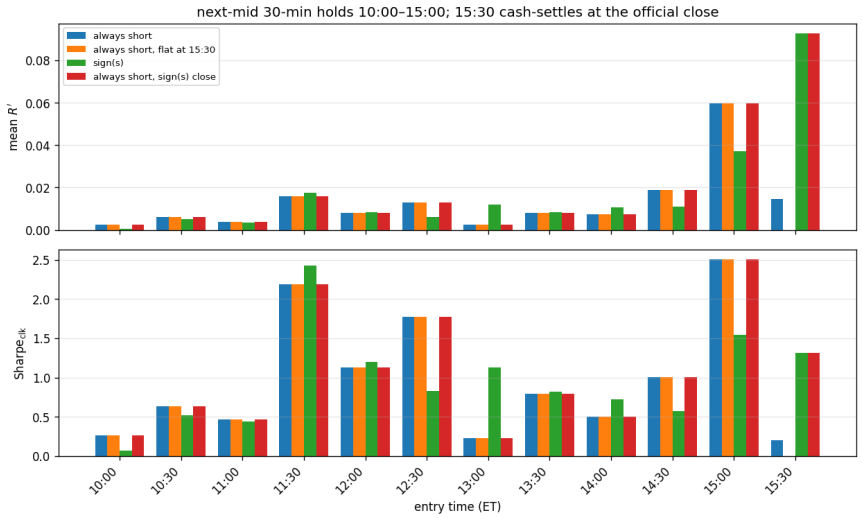
0DTE nearest-OTM straddle: per-rule return summary, pooled over the twelve entry windows 10:00–15:30 ET (daily sums), models compared on the same 866 days

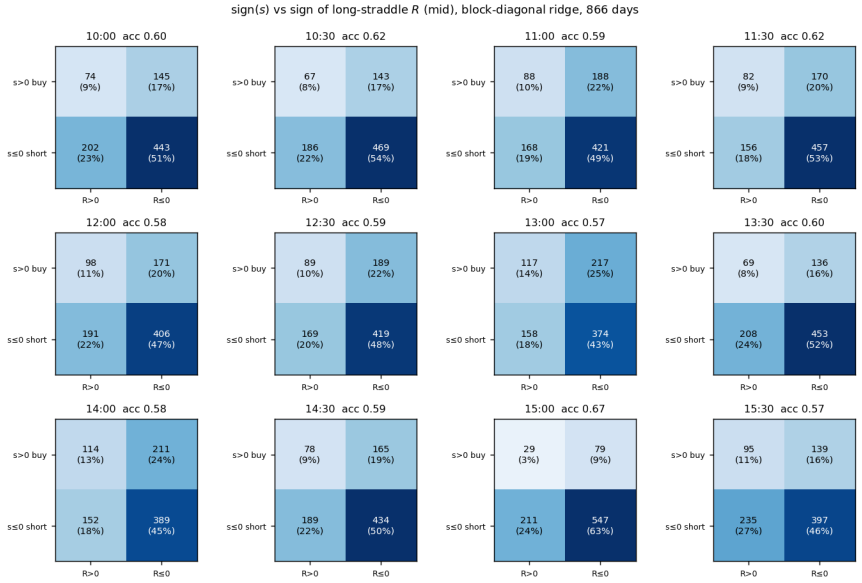
	n	mean	std	min	max	skew	ex. kurt	t	$\text{Sharpe}_{\text{ann}}$	n_{buy}	%buy	$\text{Sharpe}^{\times}_{\text{ann}}$
Short volatility (always short)												
all models (no forecast)	866	0.160	1.338	-11.40	2.24	-1.92	8.73	3.53	1.90	0	0.0	-1.92
sign(s) (*± 1 on the sign of s*)												
baseline (HAR + calendar OLS)	866	0.161	1.321	-8.94	6.22	-1.00	4.63	3.58	1.93	122	14.1	-2.36
block-diagonal ridge	866	0.212	1.320	-6.15	10.65	0.16	6.23	4.72	2.55	149	17.2	-1.72
block-diagonal ridge, without the FOMC columns	866	0.219	1.321	-6.15	10.65	0.15	6.25	4.88	2.63	157	18.1	-1.64
LightGBM†	866	0.223	1.296	-6.23	10.65	0.00	6.35	5.07	2.73	122	14.1	-1.62
XGBoost†	866	0.216	1.312	-9.98	6.95	-0.93	6.36	4.84	2.61	119	13.7	-1.71
lasso (causally tuned)†	866	0.216	1.346	-7.45	11.70	-0.14	8.51	4.73	2.55	130	15.0	-1.64
lasso ($\alpha = 10^{-4}$)	866	0.193	1.310	-6.72	10.10	-0.22	5.62	4.33	2.34	157	18.1	-1.94
elastic net (causally tuned)†	866	0.169	1.312	-7.45	11.70	-0.05	9.38	3.78	2.04	137	15.8	-2.23



Distribution of the return of the daily sum over the twelve entry windows, block-diagonal ridge forecast, one panel per rule (returns inside their 1st–99th percentile window).



Mean R' and $\text{Sharpe}_{\text{clk}}$ by entry clock (grouped bars). $\text{Sharpe}_{\text{clk}}$ is that clock's daily series $\times\sqrt{252}$, not the pooled $\text{Sharpe}_{\text{ann}}$. Next-mid holds 10:00–15:00 (25 minutes from 09:35, 30 minutes from every later clock); 15:30 cash-settles at the official close.



Confusion matrix per entry clock: sign(s) (buy if $s > 0$) against the sign of the long-straddle midpoint return. 15:30 is the last-30-min trade.

Each row is the daily sum over the twelve entry windows 10:00–15:30 ET: one number per expiration day, then the usual statistics on that daily series. 866 expiration days, 2020-01-03 to 2024-04-30 (the deck's frame). Fills are at the quoted midpoint everywhere except the last column, which is the same rule filled at the crossed spread: entry at the touch (the ask when long, the bid when short) and exit at the touch of the next stamp; the 15:30 leg cash-settles and pays no exit spread, and a re-pick that lands on the same two strikes with the same sign is a hold, not a round trip, so it is not charged. **Straddle** here means the nearest out-of-the-money call plus the nearest out-of-the-money put, same-day expiry, bought or sold as one position. One expiration day = one return; mid fill. Every t here is the plain $t = \sqrt{n} \cdot \text{mean/std}$, with no autocorrelation correction. "ex. kurt" is the bias-corrected sample excess kurtosis (Gaussian = 0). The short-volatility rule takes no forecast, so it is one row for all models. **Two Sharpes.** Both multiply by $\sqrt{252}$ because an expiration day is the unit of time (same year-length as the paper). They are not the same object. On a *single-clock* page, $\text{Sharpe}_{\text{clk}} = (\text{mean/std}) \times \sqrt{252}$ of that clock's series: one 30-minute hold per expiration day. It answers "if I only entered at this clock, what is my annual Sharpe?" It is *not* the book's $\text{Sharpe}_{\text{ann}}$. On the *pooled* page, $\text{Sharpe}_{\text{ann}} = (\text{mean/std}) \times \sqrt{252}$ of $R_d^{day} = \sum_t R'_{d,t}$, the sum of the day's twelve holds. That reserved name is the book's daily P&L after pooling the session. At 15:30 the two coincide (one trade that day); they do not on 10:00–15:00 or on the pooled page. $\text{Sharpe}^{\times}$ is the same convention at the crossed spread. Every other column is daily. n_{buy} is the number of expiration days with position > 0 (always-short is 0; on the pooled page it is days the sum of the twelve positions is positive); %buy is $100 \cdot n_{\text{buy}}/n$, the notebook's column. The other column this table adds to the deck's is $\text{Sharpe}^{\times}_{\text{ann}}$, the same rule's annualized Sharpe at the *crossed spread* instead of the midpoint; every other column is the deck's.

Forecasts: baseline (HAR + calendar OLS); block-diagonal ridge (HAR block and exogenous block, separate penalties) on the design carrying the FOMC calendar block, and the same ridge on the earlier design without those columns, as a diagnostic row; LightGBM and XGBoost on the all-features design; lasso on the same design, causally tuned vs. fixed $\alpha = 10^{-4}$; elastic net (causally tuned). † marks a column still fitted on the earlier design panel; a run of those four on the panel of record is pending. The signal is $s_t = \widehat{RV}_t - \text{IV}^2_{\text{hr}} h_t w_t$: the forecast against the implied variance of the same window. w_t is the expanding mean of each prior day's remaining-session share of realized variance, $\text{RV}_t / \sum_{s \geq t} \text{RV}_s$ (in-fit panel history back to 2001), not the ratio of trailing clock-means. On the bars whose vendor implied volatility is a censored solver node the deck's treatment is used here too — the volatility that reproduces the straddle midpoint, recovered by bisection over the remaining h_t hours — so every bar carries a signal and no bar sits flat.